Data Cleaning, Validation, and Analysis Workflow: Grocery Sales Dataset

Demonstrating structured data preparation, auditability, and insight generation using Excel

Introduction

I created a structured Excel data-cleaning and validation workflow using a grocery sales dataset (source: Kaggle).

The project began with data curation and quality control (QC) to ensure the dataset was reliable and analysis-ready. These steps are described in the **Data Cleaning & QC** section.

Once the dataset was validated, analysis proceeded using Excel pivot tables, which also highlighted the need for derived columns within the source data. These steps are described in the **Analysis** section.

Data Cleaning & QC

1) Unique Identifier ("UID")

I added a **UID** column to provide a unique identifier for each row. This preserves original row order and allows the dataset to be safely sorted, filtered, and restored.

2) Checksum

To ensure data integrity during sorting and transformation, I used a checksum approach to confirm that row-level values remained consistent.

Specifically, I validated that the aggregate total across all numeric columns remained unchanged before and after data manipulation.

This provided a quick integrity check that no rows were lost, duplicated, or altered during processing, reinforcing a "trust before analysis" approach to data handling.

3) Data Typing

I converted the **quantity** field from text to numeric format to ensure reliable calculations. The **customer_id** field was retained as text, as it functions as an identifier rather than a measurable value.

4) Excel Table Structure

The dataset was converted into an Excel Table to support structured references, consistent formulas, filtering, and scalable row-level validation.

5) Missing Value Checks

Using the Excel Table's Total Row functionality, I applied:

- **COUNT** for numeric columns
- **COUNTA** for non-numeric columns

The **UID** column confirmed a total of 1,980 rows.

All columns returned a count of 1,980 except **store_name**, which returned 1,955, indicating 25 missing values.

To isolate these records, I created a helper column (**store_name_present**) to flag each row as **"Missing"** or **"OK."** The **Missing** rows were excluded from analyses that relied on store-level grouping.

6) Categorical Field Profiling

I used **SORT(UNIQUE())** with structured table references to profile categorical fields and detect inconsistencies such as typos, case variation, or extra spacing.

The following columns were evaluated:

- **store_name**
- **aisle**
- **product_name**

No inconsistencies were identified that would impact aggregation. A complete listing of unique values is provided in Appendix B.

7) Calculated Fields QC

To validate calculated values, I created a derived column:

- **calc_final_amount = (quantity × unit_price) – discount_amount**

I then compared this to the provided **final_amount** field using a QC difference column (**qc_calc_final**).

The total difference across the dataset was effectively zero, confirming that the calculated values matched the source field within expected rounding precision.

Supporting data and validation structures are included in Appendix A, with additional categorical field profiles provided in Appendix B.

QC SUMMARY

This project demonstrates a structured data quality workflow, including:

- preserving source order
- validating data types
- identifying missing values
- profiling categorical fields
- validating calculated fields

These steps ensured the dataset was reliable and suitable for downstream analysis and reporting.

Analysis & Results

After validating the dataset, analysis was conducted using pivot tables to evaluate product distribution, revenue patterns, and store-level variation.

1) Product Volume

Total product volume by store showed some variation; however, most locations fell within a relatively narrow range (192–235), indicating broadly consistent transaction levels across stores.

Product-level differences were modest, with total counts ranging from 88 to 126 across items. Most products fell within a relatively narrow band, indicating no strong dominance by any single category.

A detailed breakdown of product volume by store and product category is provided in Appendix C (Table C-1).

2) Revenue Patterns

Revenue patterns closely mirrored product volume, suggesting that price variation did not significantly impact total sales within this dataset.

This is supported by relatively uniform unit pricing across stores and products (14–17), as well as consistent discount levels, which showed minimal variation across stores (12–13%) and only modest variation across products (11–17%).

In a more variable pricing environment, revenue patterns might diverge from volume; however, in this dataset, they remained closely aligned.

3) Derived Metric: Loyalty Points per Revenue

To evaluate whether additional variation existed beyond what was observed in the initial analysis, a derived metric representing loyalty points relative to transaction value (**loyalty_points_per_final_amount**) was created.

For aggregate analysis, loyalty points per final amount was calculated at the store level by dividing the sum of loyalty points by the sum of final amount, enabling accurate comparison across stores.

While overall product volume and revenue were relatively consistent, this derived metric showed variation across stores, indicating differences in loyalty or promotional behavior (see Figure 1 and Appendix C, Table C-2).

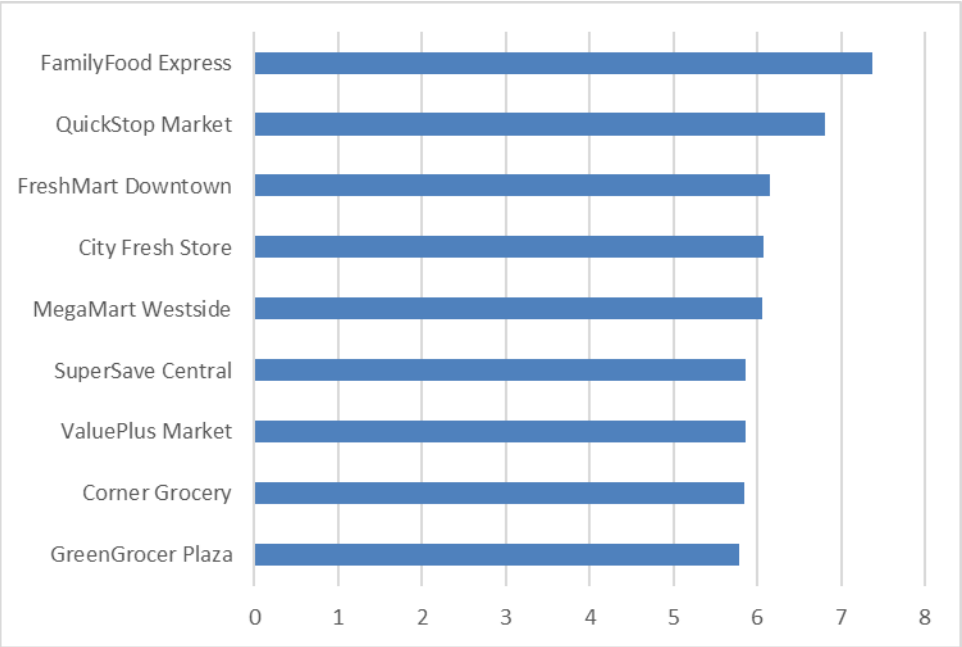

Fig. 1: Loyalty Points per Final Amount by Store

This highlights the value of derived metrics in revealing patterns not immediately visible in aggregate totals.

4) Results Summary

Product volume and revenue patterns showed relatively limited variation across stores, driven primarily by similar product volumes, consistent pricing, and relatively uniform discount structures. A derived metric (loyalty points per final amount) revealed additional variation across stores that was not apparent in aggregate totals.

The dataset and all transformation steps are included in Appendix A to provide a clear audit trail and full transparency of the data preparation process.

Project Overview

This project demonstrates a structured and validated data preparation workflow (ETL), ensuring that analysis was performed on reliable, well-defined data.

Initial analysis indicated relatively consistent patterns across stores, with limited variability in the primary measures. To further evaluate potential differences, a derived metric (loyalty points per final amount) was introduced, revealing variation across stores that was not apparent in aggregate totals.

Together, these results highlight the importance of validating data prior to analysis and using thoughtfully designed metrics to uncover meaningful patterns. The inclusion of a clear audit trail supports transparency and reinforces confidence in the analytical results.

Supporting materials and detailed outputs are provided in Appendices A, B, and C.

Appendix A – Data & QC Structure

This appendix documents the structure of the cleaned dataset and the quality control (QC) methods applied to ensure data reliability and auditability.

The primary dataset is contained in the **Appendix_A_Data_QC** worksheet within the accompanying Excel workbook (**Grocery_Data_Analysis_Workbook.xlsx**). This sheet includes the original fields along with additional helper columns used for validation and analysis.

Key QC steps included:

- Addition of a unique identifier (**UID**) to preserve row traceability during sorting and filtering
- Validation of numeric fields to ensure proper data types for calculation
- Identification of missing values using count-based checks and helper columns
- Use of checksum-style validation to confirm that aggregate numeric values remained consistent during data manipulation
- Verification of a calculated field by comparing derived values to the source field

Curation and QC with use of derived columns:

- **store_name_present** was made to find the missing values in the **store_name** column
- A derived calculation was used to validate transaction-level totals:
    - **calc_final_amount** = (**quantity** × **unit_price**) – **discount_amount**
    - **qc_calc_final** confirmed that calculated values matched the provided values (**final_amount**) within expected rounding precision.

Together, these steps establish a clear audit trail and ensure that downstream analysis is performed on reliable, well-defined data.

Appendix B – Categorical Field Profiles

This appendix provides categorical field profiling used to evaluate data consistency across key grouping variables.

Unique values for selected categorical fields were generated using structured table references with:

- SORT(UNIQUE())

The following fields were evaluated:

- **store_name**
- **aisle**
- **product_name**

These profiles are available in the **Appendix_B_QC_Profiles** worksheet within the Excel workbook.

The profiling process supports identification of potential inconsistencies such as:

- typographical errors
- case variation
- leading or trailing spaces

No inconsistencies were identified that would impact aggregation or analysis.

Appendix C – Pivot Analysis

This appendix summarizes the pivot table outputs used to support analysis and results.

Pivot tables were used to evaluate:

- product volume distribution
- derived metrics including loyalty points relative to transaction value

These outputs support the findings described in the Analysis & Results section and are available in the Excel workbook (**Appendix_C_Pivot_Analysis** worksheet).